

Collections and Generics in C#

Engineering Custom Linear Collections

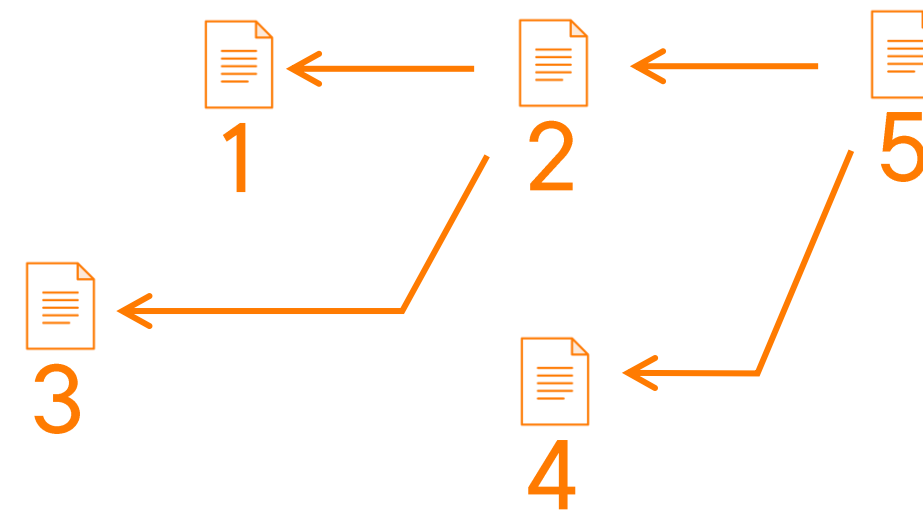
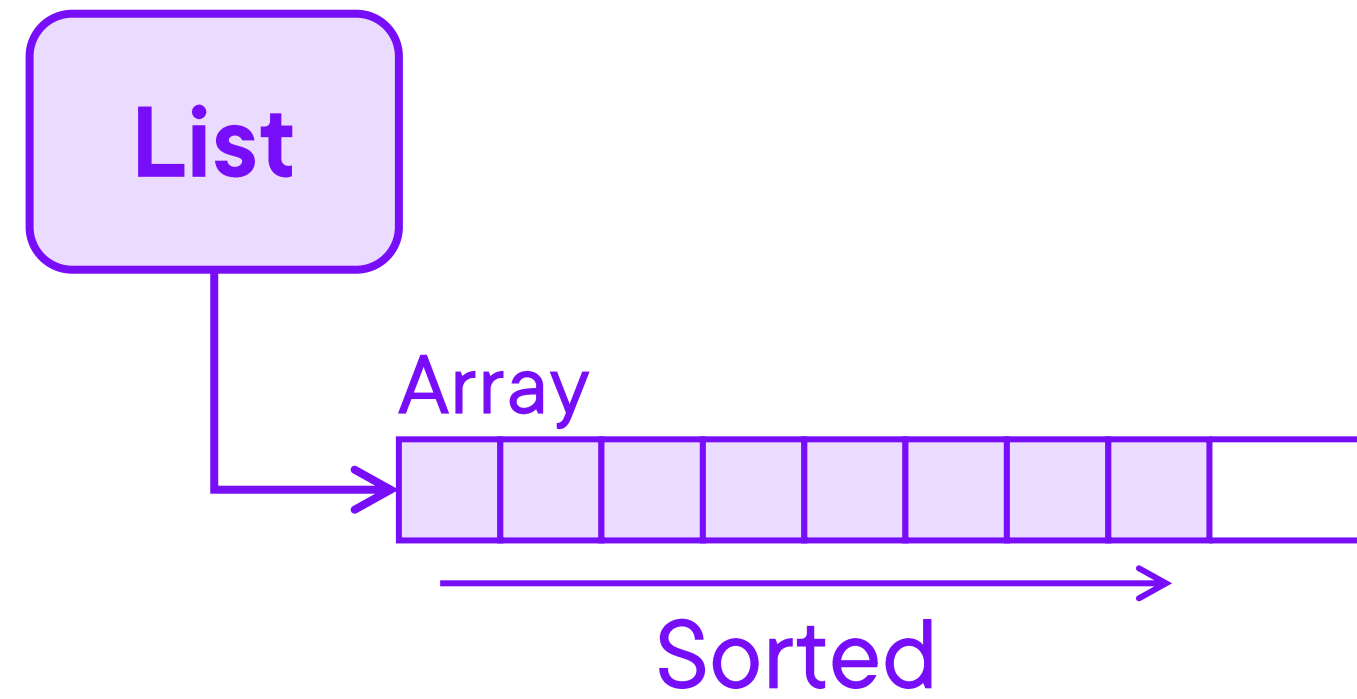


Zoran Horvat

CEO at Coding Helmet

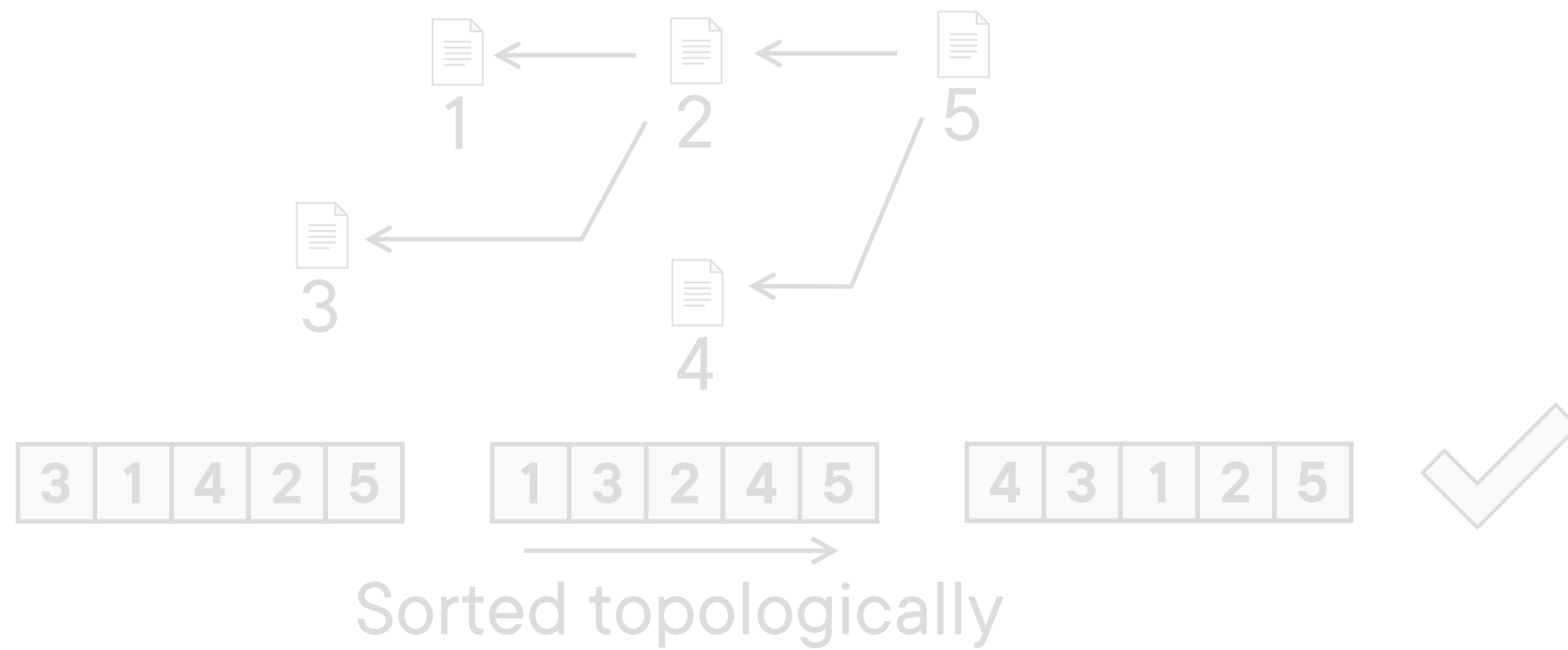
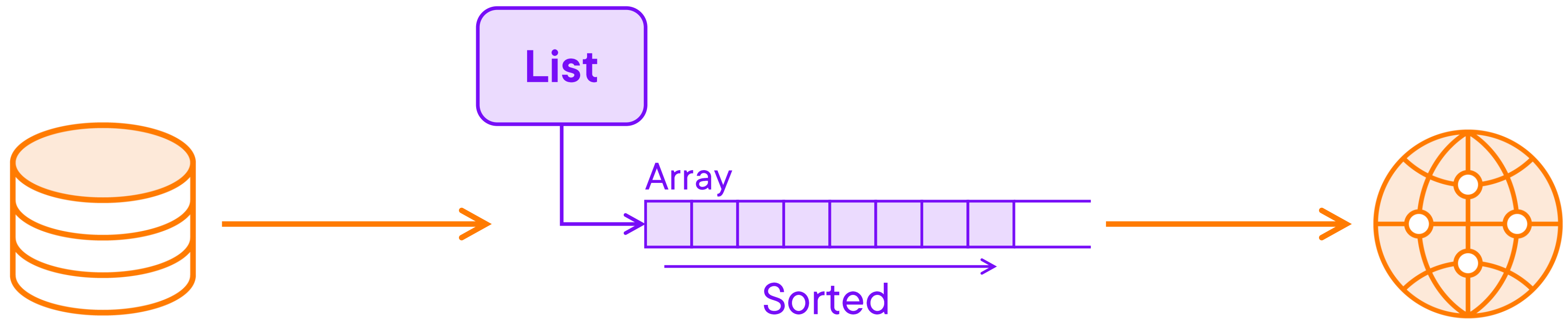
@zoranh75 | <https://codinghelmet.com>

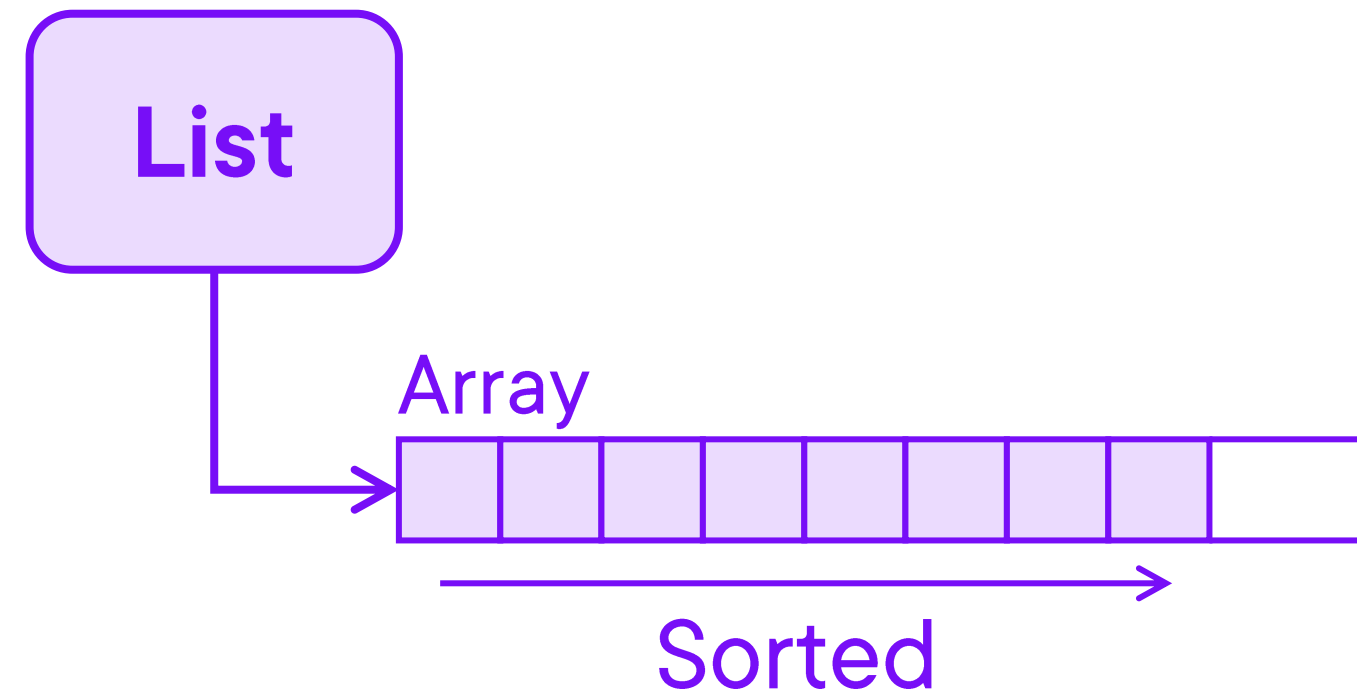




Sorted topologically

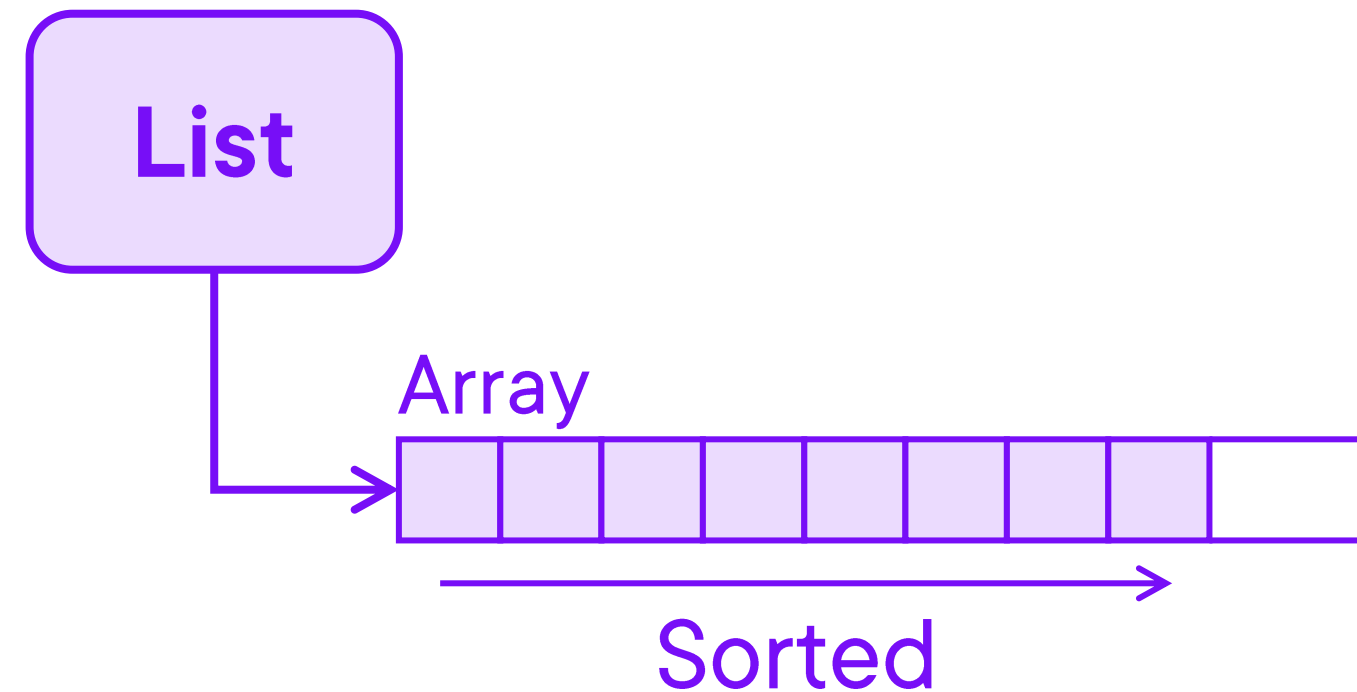






Use the `IOrderedList<T>` to calculate percentiles of a sequence

Use the `IOrderedList<T>` to remove outliers from a sequence



Customer's requirements:

Given: Sequence of objects
Sorting criterion
Percentage value

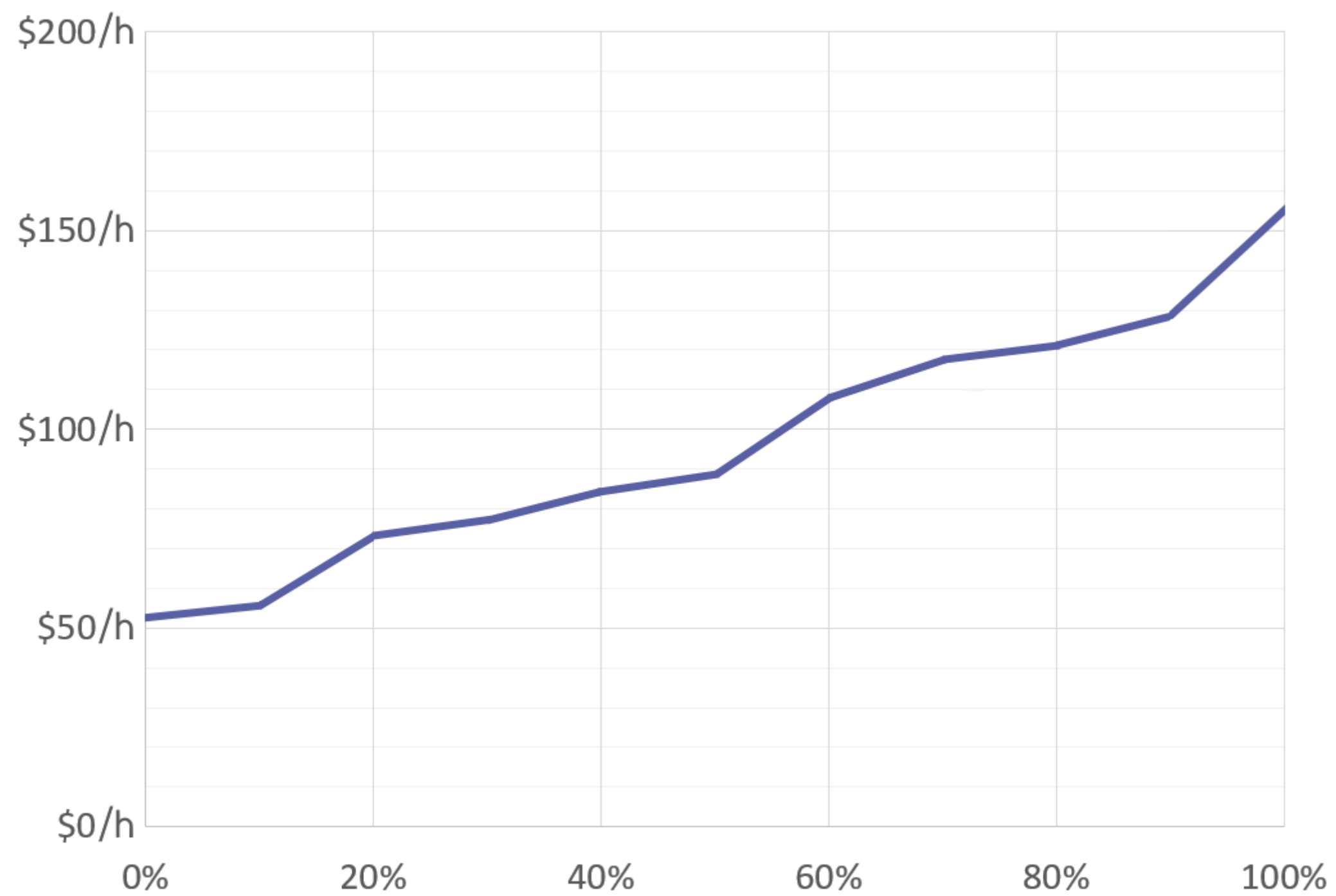
Calculate: Return the requested percentile object
(that many of other objects are less or equal to it)

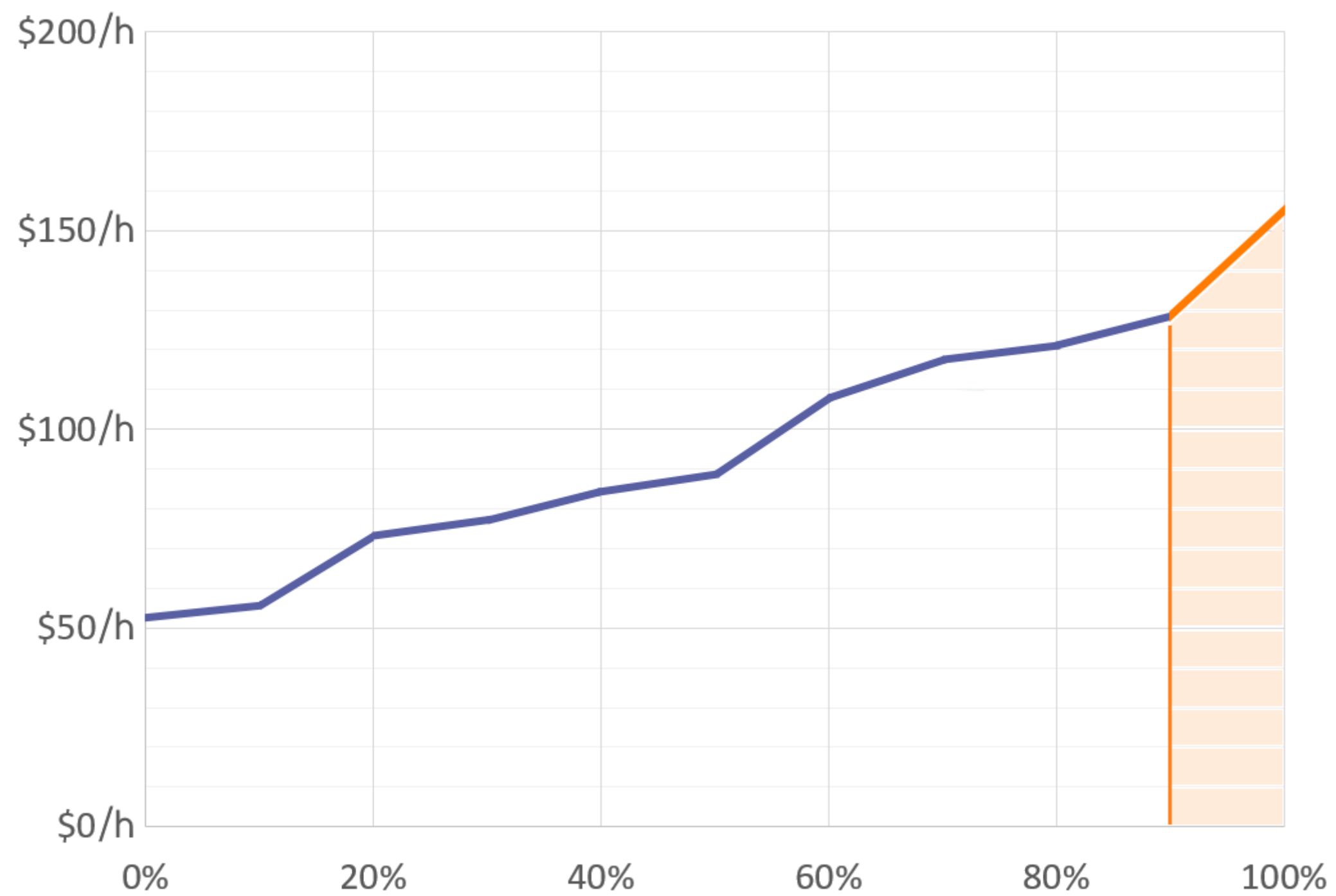
Example: Pay rate limit for lowest-paid 10%, 20%, ... of workers

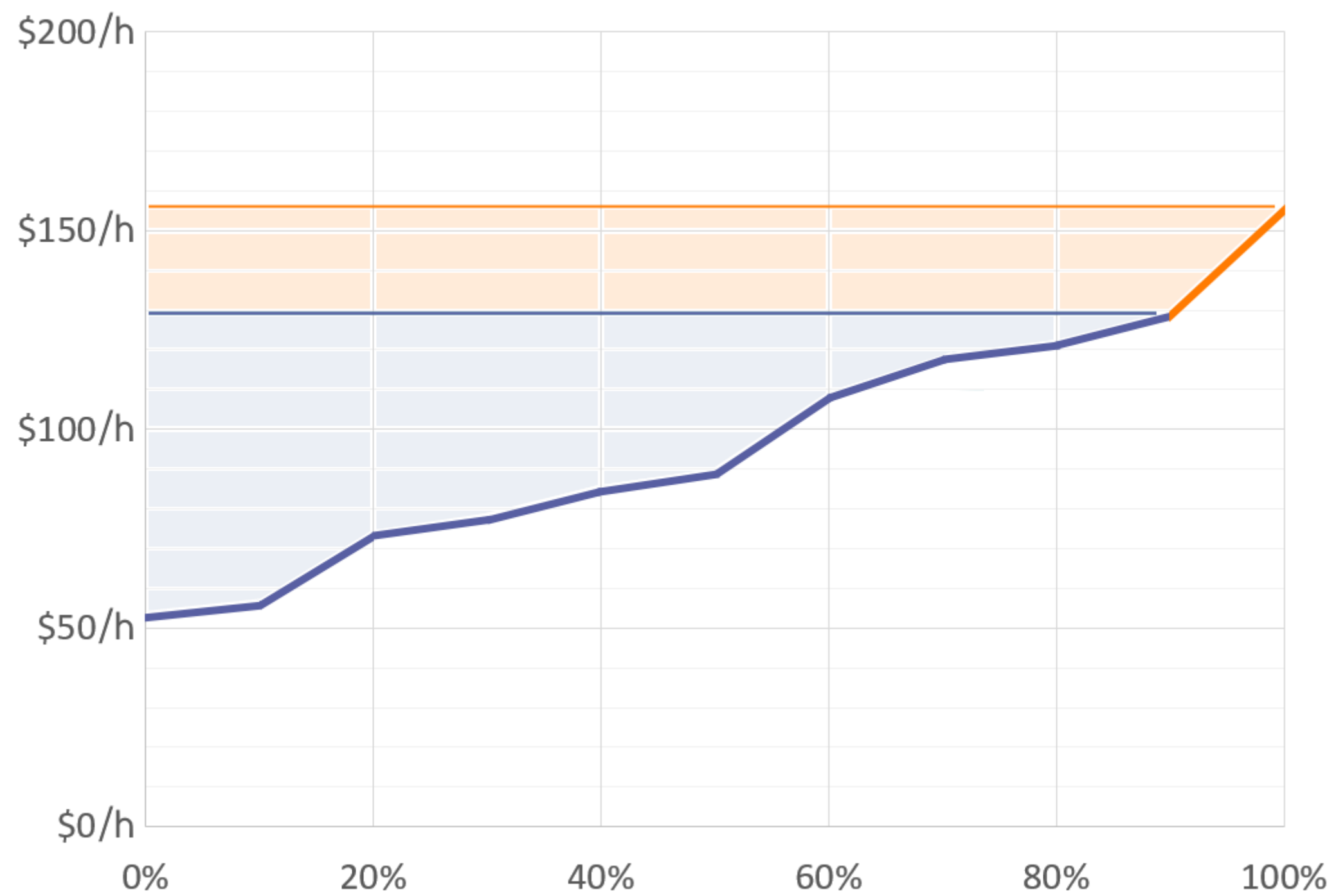
Gang of Four, 1994

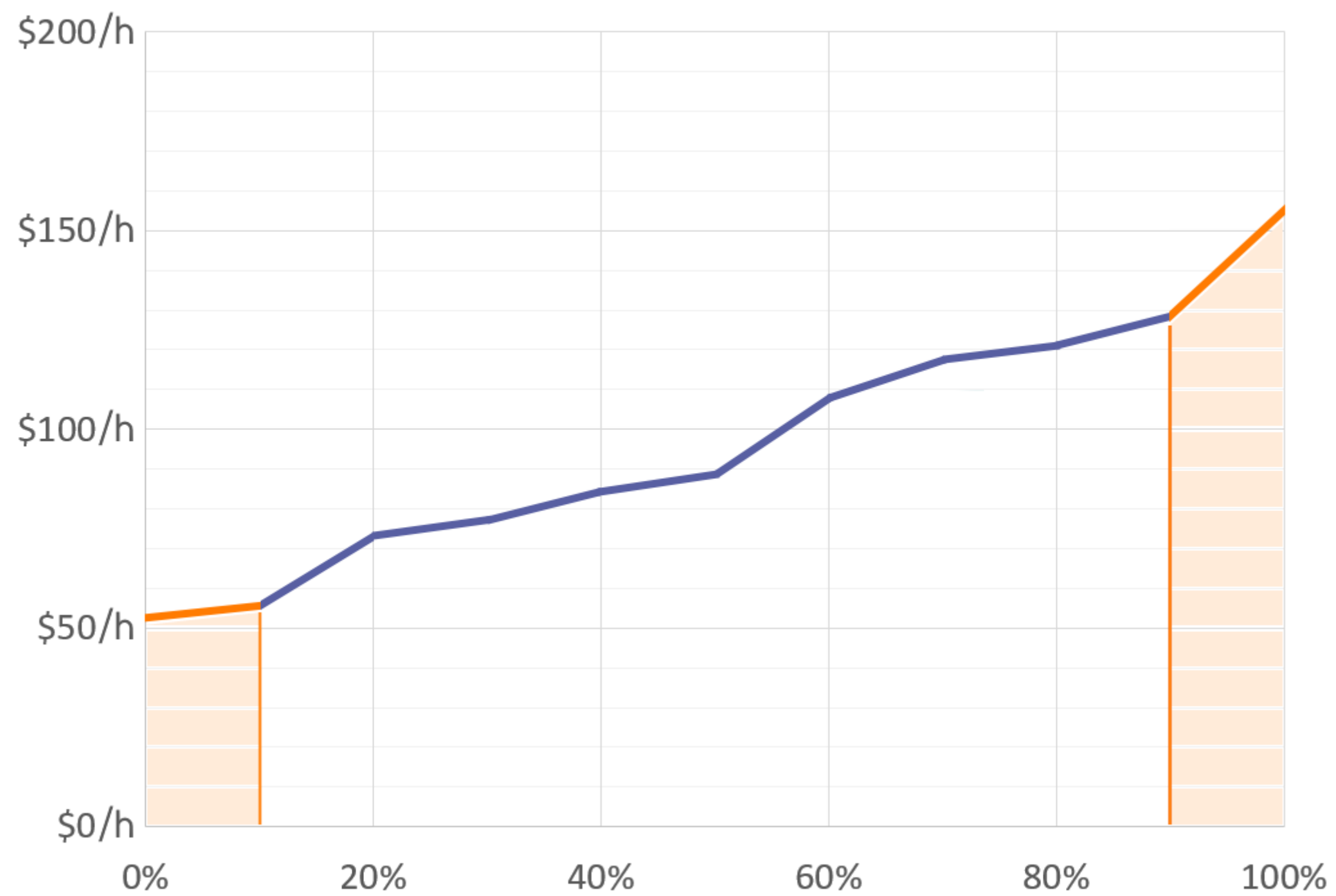
**“Favor object composition
over class inheritance.”**

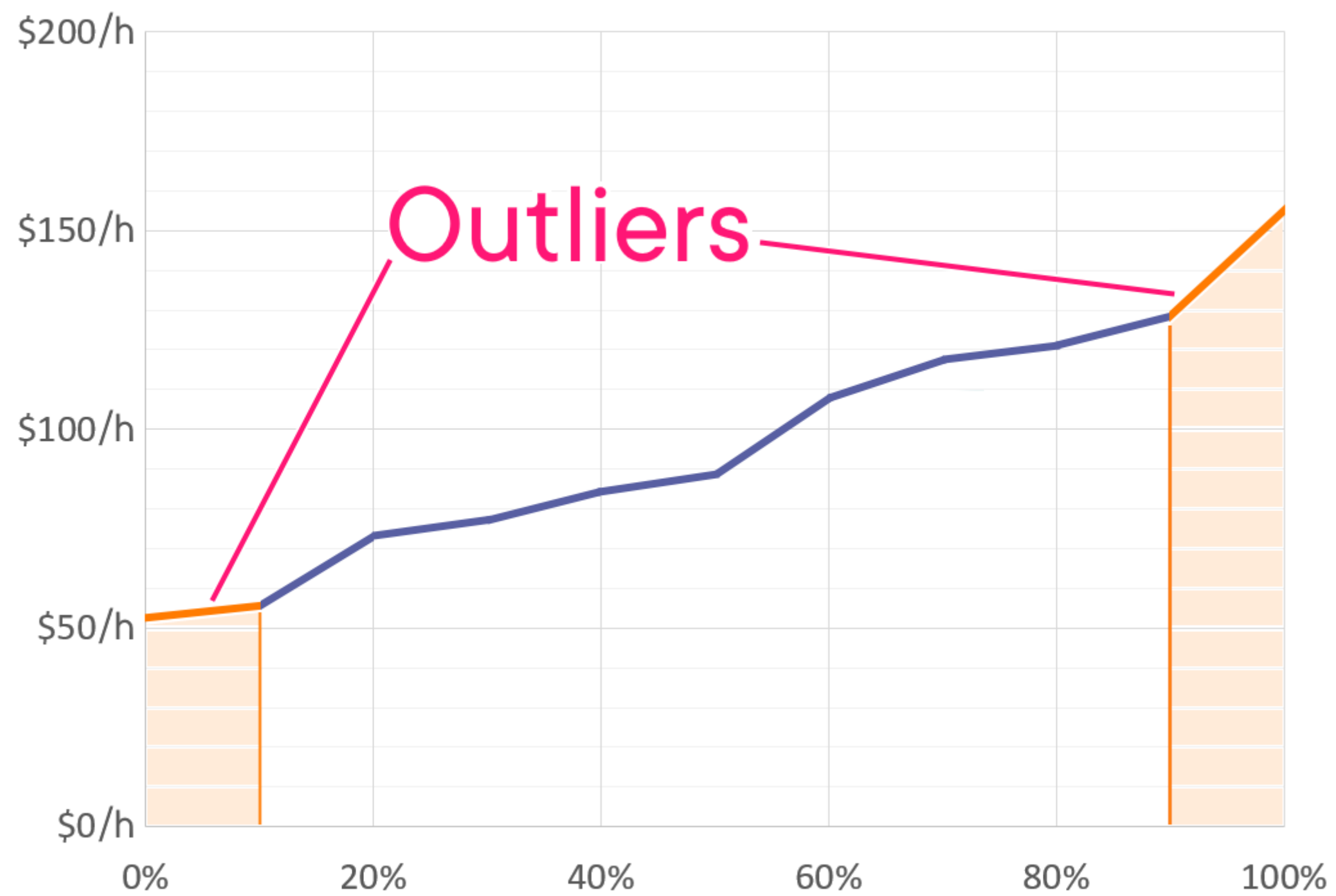












393 KB														
	PlainCreate		100000		6,483.43 us		74.638 us		69.816 us		1.00		0.00	492.1875 242.1875 242.1875
3,908 KB														
	CachedCreate		100000		8,563.27 us		101.076 us		89.601 us		1.32		0.02	484.3750 234.3750 234.3750
3,909 KB														
	PlainCreate		1000000		101,362.43 us		1,482.499 us		1,237.954 us		1.00		0.00	4400.0000 2400.0000 600.0000 3
9,066 KB														
	CachedCreate		1000000		80,267.79 us		686.799 us		608.829 us		0.79		0.01	4000.0000 1285.7143 285.7143 3
9,066 KB														

```
// * Hints *
Outliers
CurrencyCreationBenchmark.CachedCreate: Default -> 1 outlier was removed (657.53 us)
CurrencyCreationBenchmark.CachedCreate: Default -> 1 outlier was removed (8.95 ms)
CurrencyCreationBenchmark.PlainCreate: Default -> 2 outliers were removed (106.21 ms, 106.66 ms)
CurrencyCreationBenchmark.CachedCreate: Default -> 1 outlier was removed (84.33 ms)
```

```
// * Legends *
CollectionSize : Value of the 'CollectionSize' parameter
Mean           : Arithmetic mean of all measurements
Error          : Half of 99.9% confidence interval
StdDev         : Standard deviation of all measurements
Ratio          : Mean of the ratio distribution ([Current]/[Baseline])
RatioSD        : Standard deviation of the ratio distribution ([Current]/[Baseline])
Gen 0          : GC Generation 0 collects per 1000 operations
Gen 1          : GC Generation 1 collects per 1000 operations
Gen 2          : GC Generation 2 collects per 1000 operations
Allocated      : Allocated memory per single operation (managed only, inclusive, 1KB = 1024B)
1 us           : 1 Microsecond (0.000001 sec)
```

```
// * Diagnostic Output - MemoryDiagnoser *
```

Outliers
removal

Customer's requirements:

Given: Sequence of objects
Sorting criterion
Percentage value

Request: Remove the specified percentage of items
from the ends of the sort order



Summary



Demonstrated use of linear collections

- Applied a list far away from the domain
- Implemented percentiles calculator
- Implemented outliers remover

Building a hierarchy of abstractions

- Improves composability of solutions
- Enables performance optimizations



Summary



Practical guidelines

- General-purpose operations belong to infrastructure
- Infrastructure should be optimal
- Domain-related operations rely on composability
- Domain should not depend on collections



Up Next:

Engineering Queuing Solutions

